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Q1). Explain the key similarities between Power BI and Excel and discuss how these common features support data analysis and business decision making:

- Both support data import from multiple sources such as databases, CSV files, cloud services etc. enabling a complete view of business data.
- Both use Power Query for data cleaning, transformation, and shaping, ensuring high quality and consistent data for analysis.
- Both support data modeling and DAX formulas, allowing creation of relationships, calculated columns and measures for deep analysis.
- Both provide data visualization tools like charts, tables and graphs to help users easily identify trends, patterns, and insight.
- Both allow filtering, sorting and slicing of data, making analysis interactive and helping users explore different scenarios quickly.

Support Business Decision Making:

- Accurate and reliable reports.
- Trend analysis and forecasting.
- Understand performance at a glance.

- Make informed decisions.
- Strategic planning and operational improvements.

Q2) Discuss how Excel can be integrated with Power BI Desktop and the Power BI Service, and explain the benefits and practical uses of this integration in real-world reporting scenarios:

- Excel integrates with Power BI Desktop by allowing direct import of Excel tables, ranges, Power Query - cleaned data and Power Pivot Data Models including relationships and DAX measures.

- These connections stay refreshable, meaning any update made in the Excel file automatically updates the corresponding Power BI reports and dashboard.

- In the Power BI Service, Excel workbooks can be uploaded for cloud access and the Analyze in Excel features let Excel connect directly to Power BI datasets using PivotTables.

- Excel files stored in OneDrive or SharePoint automatically sync with Power BI ensuring real-time updates in dashboards without manual uploads.

- This integration offers strong benefits like combining Excel's analytical power with Power BI's interactive visuals, reducing manual

effort through auto-refresh improving Collaboration and Supporting real-world uses. Such, financial reporting, sales dashboards, inventory tracking, HR insight and executive decision making.